

PHYSICS 550
Homework #1
Due on Sept 8, 2023

1. (10 points)

a) This part of the problem demonstrates that the state of the system is completely specified by the probability density, $\rho(\vec{r}) = |\psi(\vec{r})|^2$, and the probability current, $\vec{J}(\vec{r}) = \frac{\hbar}{2im} [\psi^*(\vec{r})\nabla\psi(\vec{r}) - \nabla\psi^*(\vec{r})\psi(\vec{r})]$. Writing $\psi(\vec{r}) = \sqrt{\rho(\vec{r})}e^{i\xi(\vec{r})}$, with $\xi^*(\vec{r}) = \xi(\vec{r})$, show that $\vec{J}(\vec{r}) = \frac{\hbar}{m}\rho(\vec{r})\nabla\xi(\vec{r})$.

Hence show that if the potential energy is nonsingular for any finite $\vec{r}$ and two wave functions give the same $\rho(\vec{r})$ and $\vec{J}(\vec{r})$, then they can differ only by a physically irrelevant constant (independent of $\vec{r}$) phase factor. The same situation applies to any number of spatial dimensions.

b) Suppose a particle moves in 1-dimension under the influence of a potential $V(x)$ having the property that $V(\pm\infty) = 0$. The probability density and current for a stationary state are found to be

$$\rho(x) = \frac{N^2}{(e^{\gamma x} + e^{-\gamma x})^2}; \quad J(x) = 0$$

where N and γ are real constants. Find the form of the potential and the value of the energy.

2. G 1.8 (10 points)

3. G 1.9 (10 points)